

1. Administrative & Study Identification

Full Study Title: A Study to Investigate Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics of AZD4954 in Healthy Adult Participants With or Without Elevated Lipoprotein (a) (Lp[a]) Levels, and Participants With Dyslipidemia **Short Title/ Acronym:** AZD4954 SAD/MAD Phase 1 Study **Protocol Number:** D7300C00001 **NCT Number:** NCT06980428 **Sponsor Name:** AstraZeneca **Investigational Product:** AZD4954 (Oral small molecule Lp(a) inhibitor) **Phase of Development:** Phase 1 **Medical Monitor:** AstraZeneca Clinical Research Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of single and multiple ascending doses (SAD/MAD) of AZD4954 in healthy participants with or without elevated Lipoprotein(a) (Lp[a]) levels and participants with dyslipidemia. **Secondary Objectives:** To characterize the single dose and/or steady-state pharmacokinetics (PK) of AZD4954 following oral administration, and to evaluate its pharmacodynamics (PD) by assessing Lp(a) level reduction.

2.2 Background & Rationale Elevated Lipoprotein(a) [Lp(a)] is a recognized, genetically determined, independent risk factor for atherosclerotic cardiovascular disease (ASCVD) and major adverse cardiac events (MACE). Currently, effective oral treatments to substantially lower Lp(a) are lacking. AZD4954 is a novel oral small molecule being developed to inhibit Lp(a). This first-in-human study aims to evaluate its safety, PK, and PD profile to support its clinical development for patients with dyslipidemia and elevated Lp(a).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind (Sponsor, Investigator, and Participant blind) **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: 120 participants **Number of Sites:** Global Phase 1 Clinical Units **Estimated Study Start:** Q2 2025 **Estimated Primary Completion:** December 2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male and female participants aged 18 to 70 years with suitable veins for cannulation or repeated venipuncture. 2. **Part A (SAD)**: Healthy participants. 3. **Part B (MAD)**: Healthy participants with elevated Lp(a) ≥ 30 mg/dL; OR participants with dyslipidemia with elevated Lp(a) ≥ 70 mg/dL and low-density lipoprotein cholesterol (LDL-C) ≥ 70 mg/dL and < 190 mg/dL.

4.2 Exclusion Criteria 1. Participants with known bleeding or coagulation disorders. 2. Elevated high-sensitivity C-reactive protein (hsCRP > 3 mg/L) or prothrombin time/international normalized ratio (PT/INR) or activated partial thromboplastin time (aPTT) $> 1.25 \times$ upper limit of normal (ULN). 3. Any positive result on Screening for serum hepatitis B surface antigen (HBsAg), hepatitis B core antibody (HBcAb), hepatitis C virus (HCV) or human immunodeficiency virus (HIV).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single ascending doses (Part A) and multiple ascending doses (Part B) of AZD4954 or matching placebo, administered once daily for up to 21 days (for MAD cohorts). **Route of Administration**: Oral **Duration of Treatment**: 1 day for SAD cohorts; 21 days for MAD cohorts.

5.2 Concomitant & Prohibited Therapy Permitted: Standard allowed medications per Phase 1 unit guidelines (e.g., paracetamol/acetaminophen for analgesia). **Prohibited**: Use of any prescribed or nonprescribed medication, herbal remedies, or high-dose vitamins within 2 weeks prior to the first dose. Use of CYP enzyme-inducing drugs (e.g., St. John's Wort) within 3 weeks prior to the first dose. Previous administration of any Lp(a) inhibitor.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to -2 Informed Consent, Medical History, ECG, Labs, Lp(a) screening Baseline Day -1
Admission to clinical study site, Baseline Vitals, Randomization	Interim	Days 1 to 21 Drug Administration, PK/PD blood draws, Safety Monitoring, ECGs	End of Study Post-Dosing Follow-up safety labs, Final PK/PD assessments, Discharge/Exit

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence, severity, and causality of Adverse Events (AEs) and Serious Adverse Events (SAEs) as a measure of safety and tolerability. **Measurement Method:** Clinical assessment, physical examinations, vital signs, ECG readings, and standard central laboratory parameter monitoring.

7.2 Secondary & Exploratory Endpoints 1. Single dose and steady-state pharmacokinetics (PK) profile of AZD4954 (e.g., C_{max} , AUC). 2. Pharmacodynamics (PD): Change from baseline in $L_p(a)$ levels following repeated oral dosing.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous daily monitoring during clinical confinement via investigator assessments, laboratory testing, and ECGs. **Serious Adverse Events (SAEs):** Must be reported within 24 hours to the sponsor and regulatory bodies. **Data Monitoring Committee (DMC):** Internal Safety Review Committee (SRC) evaluates blinded safety and PK data before authorizing dose escalations to the next cohort.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Analysis Set (all participants who receive at least one dose of the investigational product); PK/PD Analysis Set (participants with evaluable concentration and biomarker data). **Sample Size Justification:** $N=120$ provides adequate characterization of safety, tolerability, and PK/PD parameters for a Phase 1 SAD/MAD study without formal hypothesis testing power calculations. **Handling of Missing Data:** Missing safety data will not be imputed. For PK/PD analysis, specific handling rules (e.g., values below the limit of quantification) will be applied per the clinical pharmacology analysis plan.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** 100% source data verification (SDV) for critical data points including informed consent, SAEs, and primary PK/PD endpoints during regular onsite CRO monitoring. **Audit Trail:** Validated 21 CFR Part 11 compliant Electronic Data Capture (EDC) system with automated audit trails and data clarification forms (DCFs).

11. Study Identifiers & Governance

Primary ID: D7300C00001 **Registry Number:** NCT06980428 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** A Study to Investigate Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics of AZD4954 in Healthy Adult Participants With or Without Elevated Lipoprotein (a) (Lp[a]) Levels, and Participants With Dyslipidemia **Official Regulatory Title:** A Study to Investigate Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics of AZD4954 in Healthy Adult Participants With or Without Elevated Lipoprotein (a) (Lp[a]) Levels, and Participants With Dyslipidemia

12. Clinical Framework (Dyslipidemia / Lp[a] Specific)

Primary Condition: Dyslipidemia / Elevated Lipoprotein(a) **Diagnosis Threshold:** Lp(a) ≥ 30 mg/dL (Healthy cohorts) or Lp(a) ≥ 70 mg/dL and LDL-C ≥ 70 mg/dL (Dyslipidemia cohorts) **Medical Methodology:** Investigational oral small molecule Lp(a) inhibitor **Optimization Target:** Lp(a) reduction, PK/PD profiling, and safety assessment

13. Study Procedures & Methodology

Management Pathway: Phase 1 clinical unit confinement and rigorous inpatient monitoring **Education Protocol (HCP):** Clinical protocol training, Investigator Brochure (IB) review, and dose-escalation procedures **Education Protocol (Patient):** Informed consent process, clinical research unit confinement rules, and AE reporting instructions **Quality Audit Mechanism:** Phase 1 unit oversight, standard CRO site monitoring, and Sponsor data verification

14. Observation & Follow-Up Schedule

Total Duration: Up to ~10 weeks per participant (including 28-day screening, up to 21-day dosing, and follow-up) **Onsite Visit Cadence:** Day -1 admission, daily continuous monitoring during the in-house dosing period, and subsequent outpatient follow-up visits **Remote Monitoring Frequency:** N/A (Inpatient Phase 1 clinical study) **Checklist Review Interval:** Daily symptom and AE review during clinical unit confinement

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				

Clinical Operations Lead | [Name] | ____ | [DD-MMM-YYYY] | | Lead
Statistician | [Name] | _____ | [DD-MMM-YYYY] |